Towards Trustworthy AI for Refurbished Product Price Prediction Using Multimodal Data

Abstract

The growing demand for affordable and sustainable technologies has strengthened the Refurbished Product (RFP) market. It not only saves costs but also supports recycling and the circular economy. Moreover, determining the price of an RFP remains inherently uncertain due to heterogeneous conditions, incomplete records, and ambiguous evidence from images and textual descriptions. Prior studies have employed various pricing models that rely on regressed price or point predictions. They fail to quantify uncertainty and often yield unreliable, overconfident prices, which reduces trust in the RFP market. This paper proposes a trustworthy, multimodal interval-prediction framework for estimating RFP prices from images and textual descriptions. The framework learns lower and upper price bounds by optimising an interval loss that explicitly models predictive uncertainty. To ensure that the predicted intervals are reliable and informative, we incorporate a tightness constraint that minimises the interval width while preserving coverage of the actual prices. We leverage pretrained CLIP ViT-L/14 encoders to extract aligned 768-dimensional image and text representations, which are fused for interval estimation. Next, due to the unavailability of an RFP dataset, we collect a large-scale dataset of 51,783 refurbished products, each accompanied by images and textual descriptions. We evaluate uncertainty quality using the PICP, MPIW, and CWC. Experimental results demonstrate that our approach yields significantly tighter and more accurate price intervals. By providing interpretable price ranges, our framework enables risk-aware purchasing and improves transparency for trustworthy AI in RFP pricing.

1 Introduction

Refurbished Product (RFP) marketplaces have emerged as a cost-effective and sustainable alternative to purchasing new devices, offering consumers access to electronics, vehicles, and household appliances at reduced prices [Sluijterman *et al.*2024]. However, pricing refurbished products remains challenging due to substantial uncertainty arising from heterogeneous product conditions, incomplete refurbishment histories, and noisy or ambiguous multimodal information, such as product images and textual descriptions [Lai *et al.*2021]. Unlike new products with standardized specifications, refurbished items often exhibit large variations in wear, repair quality, and remaining lifespan, making accurate and reliable pricing difficult [Fathalla *et al.*2020, Han *et al.*2021].

Figure 1: Illustration of an example scenario demonstrating how trust can be established using the proposed interval-based framework, in contrast to point-based price prediction in the RFPs.

Most existing approaches [Bergmann and Feuerriegel2025, Jin2021] to automated product pricing, including recent multimodal learning-based methods, formulate the task as a point-estimation problem, predicting a single price for each product. While such models may achieve low average error, they fail to quantify uncertainty and frequently produce overconfident predictions, particularly in the presence of data sparsity, outliers, or distributional shift. In real-world refurbished marketplaces, these limitations reduce user trust and can lead to poor decision-making for both buyers and sellers, as a single predicted price provides no indication of reliability or risk [Sluijterman *et al.*2024].

From a practical and trustworthiness perspective, price interval prediction offers a more suitable alternative. By predicting a range of plausible prices rather than a single

value, interval-based models explicitly communicate uncertainty and enable risk-aware decision-making. For buyers, price intervals convey confidence in the valuation; for sellers and platforms, they provide a mechanism to manage pricing risk and market volatility. Despite these advantages, interval prediction has received limited attention in multimodal pricing literature, particularly in the context of refurbished products, where uncertainty is inherent and unavoidable. Figure 1 presents an example scenario highlighting the limitations of point-based price prediction in the RFP and demonstrates how the proposed interval-based framework establishes trust.

In this work, we propose **a trustworthy multimodal price interval prediction framework for refurbished products**. Our approach jointly **leverages visual and textual information** by integrating pretrained multimodal representations and learning lower and upper bounds of product prices at a target confidence level. The framework is designed to produce intervals that are well-calibrated and informative, balancing high coverage of true prices with minimal interval width. To further address the heterogeneity of refurbished markets, we adopt a category-aware modeling strategy, which is essential for achieving reliable calibration.

This work makes the following key contributions:

- **RFP Multimodal Dataset.** We construct a large-scale dataset of **51,783 RFP listings** spanning **13 product categories**, where each instance is associated with a product image, a textual description, and a listed price.

- **Multimodal Price Interval Prediction Framework.** We propose a trustworthy multimodal learning framework that jointly leverages visual and textual information to predict **price intervals**, explicitly modeling the inherent uncertainty in RFP pricing.

- **Uncertainty-Aware Learning Objective.** The proposed framework optimizes an interval prediction objective that balances empirical coverage and interval tightness, producing price ranges that are both **well-calibrated and informative**.

- **Category-Aware Modeling for Heterogeneous Markets.** To address extreme scale differences and heteroscedasticity across RFP categories, we adopt a **category-wise training strategy**, which significantly improves calibration consistency and interval reliability.

- **Comprehensive Evaluation and Practical Insights.** We conduct extensive experiments using standard uncertainty metrics (PICP, MPIW, and CWC) and provide qualitative case studies demonstrating that the predicted price intervals are interpretable and support **risk-aware decision-making** in real-world.

Paper organization: Section 2 presents the description of prior work. Section 3 introduces the RFP dataset. Section 4 discusses the proposed approach, followed by the experimental setup in Section 5. Section 6 evaluates the performance and Section 7 concludes the paper.

2 Related Work

Price Prediction for RFP

Price prediction for second-hand and RFPs has received increasing attention due to uncertainties arising from variable product conditions, incomplete seller information, and dynamic market behavior. Early deep learning approaches focused on end-to-end price prediction for second-hand items using visual cues extracted from product images [Fathalla *et al.*2020]. This line of work was extended through price suggestion systems for online second-hand marketplaces, which incorporated historical transaction data and textual descriptions to improve prediction accuracy [Han *et al.*2021].

More recent studies adopted multimodal learning to better capture heterogeneous information in product listings, showing that combining textual and visual features yields superior performance compared to unimodal models [Han *et al.*2020]. Vision-based pricing approaches further demonstrated that product images alone can encode meaningful pricing signals [Chen *et al.*2018]. Beyond second-hand markets, price prediction has also been explored in broader e-commerce and used-car domains, where machine learning models address challenges such as price dispersion and heteroscedasticity [Jiang2024, Hankar *et al.*2022]. However, most existing methods focus on point estimation and do not explicitly model predictive uncertainty, limiting their reliability for real-world RFP pricing.

Multimodal Price Prediction and Feature Fusion

To better exploit heterogeneous product information, multimodal learning has been widely adopted for price prediction tasks. Several works integrate visual, textual, and categorical features within unified deep learning frameworks, demonstrating that multimodal fusion consistently outperforms unimodal models in retail price estimation [Redwan *et al.*2025]. Generic multimodal pricing approaches further confirm the effectiveness of combining image and text representations for regression-based pricing, while adaptive multimodal models in related domains such as financial markets highlight the robustness benefits of leveraging diverse data sources [Zehtab-Salmasi *et al.*2023, Zehtab-Salmasi *et al.*2023]. However, these methods largely focus on point prediction objectives.

From a representation learning perspective, large-scale vision–language pretraining has emerged as a powerful foundation for multimodal applications. Models trained to align visual and textual representations with natural-language supervision exhibit strong transferability across downstream tasks [Radford *et al.*2021]. Surveys on vision–language models and image–text multimodal learning document rapid progress in multimodal representation and fusion strategies [Zhang *et al.*2024b, Guo *et al.*2024]. Despite these advances, most multimodal pricing models do not exploit pretrained vision–language representations for uncertainty-aware prediction, leaving a critical gap unaddressed.

Multimodal Systems, Trustworthiness, and Deployment

As multimodal systems are increasingly deployed in real-world applications, issues of efficiency, reliability, and trustworthiness have become central. Recent surveys emphasize architectural designs for efficient multimodal large language

models that balance performance and computational cost for scalable deployment [Jin *et al.*2025], while broader reviews highlight both the reasoning capabilities and robustness challenges of vision–language models [Zhang *et al.*2024a]. Trust and safety concerns are particularly critical in high-stakes settings, with studies revealing vulnerabilities arising from cross-modal interactions and the need for reliable and interpretable outputs [Liu *et al.*2024]. Complementary deployment-centric research further underscores that practical multimodal systems must be accurate, transparent, and well-calibrated to distribution shifts to ensure trustworthy operation [Liu *et al.*2025].

3 Dataset and Exploratory Analysis

3.1 Dataset Description

In this work, we compile a large dataset of RFP listings from online marketplaces. Each listing includes a **product image**, a seller-provided **textual description**, and the listed price (Figure 4). After preprocessing and data cleaning, the final dataset contains **51,783 RFP samples** spanning **13 distinct product categories**, including consumer electronics, vehicles, household appliances, books, and real estate. This diversity enables, for the first time, a comprehensive evaluation of pricing behaviour across heterogeneous RFP domains.

Further, we witnessed a high level of class imbalance in the dataset. High-frequency categories include laptops, bikes, fridges, televisions, and washing machines, while categories such as games and cycles contain relatively fewer samples. This imbalance reflects real-world refurbished markets, where certain product types are more prevalent, and poses additional challenges for learning robust and well-calibrated pricing models. Each data instance contains multimodal inputs: (i) an image capturing the visual condition of the product, and (ii) a textual description summarizing specifications, usage history, and refurbishment details. The target variable is the listed resale price, expressed in Indian Rupees.

Table 1: An illustration of collected dataset statistics. All mentioned prices are in rupees.

Category	Samples	Min Price	Max Price	Avg. Price
Bikes	6190	15K	1550K	107K
Books	3020	45	15K	1774
Cars	4400	66K	11950K	968K
Cycle	1910	896	800K	13K
Flat	3370	2597K	40000K	7357K
Fridges	5000	3199	150K	11K
Games	120	1800	40K	21K
Game Ent.	4970	12K	100K	16K
Laptop	8350	16K	250K	30K
Mobile	6550	2200	138K	28K
Printer	2630	1940	185K	30K
TV	4800	6499	455K	16K
WM	473	5210	200K	8848
Total	**51783**	–	–	–

3.2 Statistical Characteristics of Prices

An exploratory analysis of raw price values reveals substantial variability and heavy-tailed distributions across product categories. High-value categories such as flats and cars exhibit extremely wide price ranges, with prices spanning several orders of magnitude. while car prices also exhibit significant dispersion (Table 1). Such behaviour results in very high standard deviations relative to median prices, indicating strong right skewness and the presence of extreme outliers.

Consumer electronics categories, including laptops, mobiles, phones, televisions, fridges, and washing machines, also display notable price variability. Although their mean prices are lower than those of vehicles or real estate, their standard deviations often exceed their medians, reflecting heterogeneity in brand, specifications, usage duration, and refurbishment quality. Lower-priced categories such as books and games, despite having smaller absolute price ranges, still exhibit right-skewed distributions due to rare high-value items.

These observations highlight that the price distributions across categories are highly non-Gaussian, heteroscedastic, and dominated by outliers, making point price prediction particularly challenging.

3.3 Log-Price Transformation

An exploratory analysis of the raw resale prices reveals substantial skewness, heavy-tailed distributions, and large-scale differences across product categories. High-value categories such as cars and flats span several orders of magnitude, while lower-priced categories such as books and games still exhibit pronounced right-skewness due to rare high-value instances. This heteroscedasticity poses significant challenges for learning reliable, well-calibrated pricing models, particularly when optimizing interval-based objectives.

To better understand these characteristics, we analyze prices in the logarithmic domain (Table 2). The log-price transformation substantially reduces variance, compresses extreme values, and yields more stable, comparable distributions across heterogeneous categories. In particular, the standard deviations of log-transformed prices are markedly smaller and more consistent than those observed in the raw price space, mitigating the dominance of high-value outliers and improving numerical stability.

Such stabilization is especially beneficial for uncertainty-aware modeling, as interval prediction methods are sensitive to extreme values and scale disparities. Log-scale analysis provides insight into the intrinsic uncertainty structure of refurbished markets and motivates the need for category-aware modeling and careful interval calibration.

Unless otherwise stated, all price interval predictions and evaluation metrics reported in this work are computed in the original price space. The log-price transformation is used exclusively for exploratory analysis and statistical characterization, and not as the prediction target, ensuring that predicted price intervals remain directly interpretable and actionable in real-world marketplace settings.

3.4 Category-wise Modeling

The exploratory analysis reveals pronounced differences in price scales, variances, and distributional characteristics

Table 2: Few samples from the per-category price statistics in raw and log-transformed domains.

Category	Mean Price	Mean Log-Price
Books	1,774	7.48
Games	21,000	9.95
Bikes	107,000	11.58
Cars	968,000	13.78
Flats	7,357,000	15.81

across product categories. Training a single global pricing model across all categories would require simultaneously modeling products with vastly different pricing mechanisms, ranging from low-cost books to high-value real estate. Such a setting often leads to poor calibration, scale dominance by high-value categories, and unreliable uncertainty estimates for lower-priced items.

To address these challenges, we adopt a category-aware modeling strategy in which separate models are trained independently for each product category. This approach constrains each model to a more homogeneous data distribution, allowing it to capture category-specific pricing dynamics and uncertainty patterns. As a result, category-wise training improves both the reliability and sharpness of the predicted price intervals, which is critical for trustworthy pricing in real-world refurbished marketplaces.

4 Methodology

4.1 Problem Formulation

Let each RFP listing be represented as a triplet $(x_i^{img}, x_i^{txt}, y_i)$, where x_i^{img} denotes the product image, x_i^{txt} denotes the corresponding textual description, and $y_i \in \mathbb{R}^+$ represents the ground-truth resale price. Instead of predicting a single point estimate, our objective is to predict a price interval $[\hat{y}_i^l, \hat{y}_i^u]$ such that the true price y_i lies within the interval with a predefined confidence level (95%).

Formally, we aim to learn a function

$$f_\theta : (x_i^{img}, x_i^{txt}) \rightarrow (\hat{y}_i^l, \hat{y}_i^u),$$

where $\hat{y}_i^l$ and $\hat{y}_i^u$ denote the lower and upper bounds of the predicted price interval, respectively.

4.2 Multimodal Feature Extraction

To jointly encode visual and textual information, we employ the CLIP-ViT-L/14 model as a pretrained multimodal backbone. CLIP is chosen due to its strong vision-language alignment learned from large-scale image-text pairs, making it well suited for heterogeneous and noisy RFP data.

Given an input image x_i^{img}, the CLIP image encoder produces a 768-dimensional embedding:

$$\mathbf{e}_i^{img} = \text{CLIP}_{img}(x_i^{img}) \in \mathbb{R}^{768}.$$

Similarly, the textual description x_i^{txt} is encoded using the CLIP text encoder to obtain:

$$\mathbf{e}_i^{txt} = \text{CLIP}_{txt}(x_i^{txt}) \in \mathbb{R}^{768}.$$

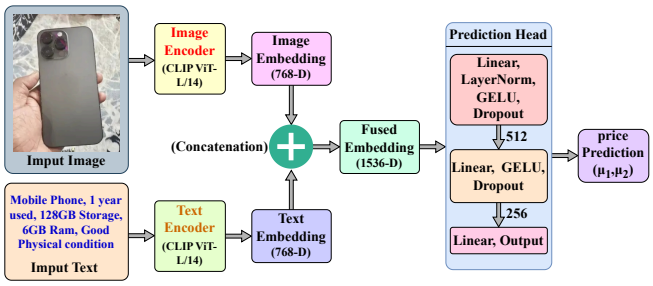

Figure 2: Proposed multimodal price interval prediction architecture.

Both encoders are kept frozen during training to ensures stable multimodal alignment and reduce overfitting, particularly for categories with limited training samples.

4.3 Multimodal Feature Fusion

The extracted image and text embeddings are fused using feature-level concatenation, resulting in a unified multimodal representation:

$$\mathbf{e}_i^{fused} = \left[\mathbf{e}_i^{img} \,;\, \mathbf{e}_i^{txt}\right] \in \mathbb{R}^{1536}.$$

This fusion strategy preserves complementary information from both modalities while maintaining architectural simplicity and interpretability. The resulting 1536-dimensional fused embedding serves as the input to the interval prediction head.

4.4 Interval Prediction Head

The fused multimodal representation $\mathbf{e}_i^{\text{fused}} \in \mathbb{R}^{1536}$, obtained by concatenating the image and text embeddings, is passed to a dedicated interval prediction head designed to explicitly model predictive uncertainty. Instead of producing a single point estimate, the prediction head outputs two scalar values corresponding to the lower and upper bounds of the price.

The interval prediction head is implemented as a multi-layer fully connected network with progressive dimensionality reduction. Specifically, the 1536-dimensional fused embedding is first projected to a 512-dimensional hidden representation using a linear layer, followed by Layer Normalization, GELU activation, and dropout to stabilize training and improve generalization. This is followed by a second linear layer that reduces the representation to 256 dimensions, again coupled with GELU activation and dropout. Finally, a linear output layer maps the 256-dimensional representation to two scalar outputs, representing the predicted lower bound $\hat{y}_i^l$ and upper bound $\hat{y}_i^u$ of the price interval.

The interval prediction head implements the function

$$(\hat{y}_i^l, \hat{y}_i^u) = f_\theta(\mathbf{e}_i^{\text{fused}}), \tag{1}$$

where f_θ denotes the learnable parameters of the prediction head. The use of non-linear activations enables the model to capture complex interactions between visual and textual features, while regularization through dropout and normalization mitigates overfitting, particularly for categories with limited training samples.

By decoupling interval estimation from the frozen multimodal encoders, the prediction head focuses on uncertainty modeling while leveraging robust pretrained representations. This design allows the model to learn asymmetric and category-specific price intervals, resulting in reliable and informative uncertainty estimates suitable for real-world RFP pricing.

4.5 Category-wise Model Training

RFP categories exhibit substantially different price scales, variances, and distributional characteristics. To address this heterogeneity, we adopt a category-aware training strategy in which separate models are trained independently for each product category.

By constraining each model to a more homogeneous data distribution, category-wise training allows the prediction head to better capture category-specific pricing dynamics and uncertainty patterns. This design improves interval calibration and prevents dominance of high-value categories over lower-priced ones, resulting in more reliable and informative price intervals across all product types.

5 Experimental Setup

5.1 Data Splitting Strategy

For each product category, the dataset is split into training, validation, and test sets using an 70:15:15 ratio. The splits are performed independently within each category to preserve category-specific price distributions and avoid data leakage across categories. All reported results are obtained on the held-out test sets, while the validation set is used for hyperparameter tuning and early stopping.

5.2 Evaluation Metrics

The quality of the predicted price intervals is evaluated using three standard metrics that jointly assess reliability and informativeness: Prediction Interval Coverage Probability (PICP), Mean Prediction Interval Width (MPIW), and Coverage Width–based Criterion (CWC) [Sluijterman *et al.*2024].

Prediction Interval Coverage Probability (PICP). PICP measures the empirical coverage of the prediction intervals, i.e., the proportion of test samples whose true prices lie within the predicted interval. Let N denote the number of test samples, and let y_i be the true price of the i-th sample with predicted interval $[\hat{y}_i^l, \hat{y}_i^u]$. PICP is defined as:

$$\text{PICP} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(\hat{y}_i^l \leq y_i \leq \hat{y}_i^u\right),$$

where $\mathbb{I}(\cdot)$ is the indicator function. A well-calibrated model should achieve a PICP close to the target confidence level (95%).

Mean Prediction Interval Width (MPIW). MPIW measures the average width of the predicted intervals and reflects their sharpness. It is defined as:

$$\text{MPIW} = \frac{1}{N} \sum_{i=1}^{N} \left(\hat{y}_i^u - \hat{y}_i^l\right).$$

Smaller MPIW values indicate tighter and more informative intervals; however, overly narrow intervals may result in insufficient coverage, highlighting the trade-off between sharpness and reliability.

Coverage Width–based Criterion (CWC). CWC combines both coverage and interval width into a single metric by penalizing intervals that fail to achieve the desired coverage level. Let μ denote the target coverage level (set to 0.95 in our experiments). CWC is defined as:

$$\text{CWC} = \text{MPIW}\left(1 + \gamma \cdot \exp\left(-\eta(\text{PICP} - \mu)\right)\right),$$

where γ is an indicator defined as:

$$\gamma = \begin{cases} 1, & \text{if PICP} < \mu, \\ 0, & \text{otherwise,} \end{cases}$$

and η is a non-negative penalty parameter that controls the severity of the coverage penalty. Lower CWC values indicate better overall interval quality, as they correspond to narrow intervals that also satisfy the desired coverage constraint.

5.3 Implementation Details

The proposed framework is implemented using PyTorch. The CLIP-ViT-L/14 image and text encoders are used as frozen feature extractors to obtain multimodal embeddings, ensuring stable representation learning and reducing the risk of overfitting, particularly for categories with limited data. All model parameters outside the CLIP backbone are trained from scratch.

For each product category, a separate interval prediction model is trained independently. The fused multimodal embeddings are passed to a fully connected prediction head, which is optimized using the Adam optimizer. The learning rate is selected based on validation performance, and early stopping is employed to prevent overfitting. Dropout and layer normalization are applied within the prediction head to improve generalization and training stability.

The dataset is split into training, validation, and test sets using an 70:15:15 ratio for each category. All reported results are obtained on the held-out test sets. To ensure robustness, model performance is averaged over multiple training runs with different random seeds. Experiments are conducted on a single GPU, and all evaluation metrics are computed at a target confidence level of 95%.

6 Results and Discussion

6.1 Performance Results

We evaluate the proposed multimodal price interval prediction framework at a target confidence level of 95% using three complementary metrics: Prediction Interval Coverage Probability (PICP), Mean Prediction Interval Width (MPIW), and Coverage Width-based Criterion (CWC). While PICP measures the empirical reliability of the predicted intervals, MPIW reflects their sharpness, and CWC jointly captures the trade-off between coverage and interval width by penalizing under-coverage.

Table 3 further presents category-wise interval prediction results for the proposed model. Across all categories, the

Table 3: Performance of the proposed multimodal interval prediction on different evaluation matrices.

Category	PICP	MPIW	CWC
Bikes	0.9570	209,307.64	209,307.64
Books	0.9581	9,345.31	9,345.31
Cars	0.9756	3,222,647.25	3,222,647.25
Cycle	0.9655	115,418.27	115,418.27
Flat	0.9671	22,194,196.00	22,194,196.00
Fridges	0.9667	33,373.81	33,373.81
Games	0.9584	37,483.30	37,483.30
Games Ent	0.9600	51,257.56	51,257.56
Laptop	0.9762	74,093.16	74,093.16
Mobile	0.9644	84,001.84	84,001.84
Printer	0.9550	10,275.00	10,275.00
TV	0.9620	62,500.00	62,500.00
WM	0.9685	42,200.00	42,200.00

model maintains PICP values at or above the desired 95% level, indicating well-calibrated prediction intervals. Importantly, interval widths scale appropriately with category-specific price ranges, yielding narrow intervals for low-priced products and wider intervals for high-value categories such as cars and flats. Since all categories satisfy the coverage constraint, CWC reduces to MPIW, confirming that the reported intervals are both reliable and informative.

6.2 Ablation Study

To determine the impact of multimodal fusion, we compare the proposed multimodal model against unimodal baselines using only visual features (image-only) and only textual features (text-only). The comparison is summarized in Table 4, which reports overall interval prediction performance aggregated across all product categories.

Table 4: Ablation study on interval prediction performance.

Model	PICP ↑	MPIW ↓	CWC ↓
Image-only	0.938	178,500	503,700
Text-only	0.945	152,300	347,900
Image + Text	**0.958**	**121,700**	**121,700**

Table 4 reports the overall interval prediction performance aggregated across all product categories for three model variants: image-only, text-only, and the proposed multimodal model. The image-only model achieves a PICP of 0.938, indicating slight under-coverage, which can be attributed to the inherent ambiguity of visual cues in RFPs where critical pricing information, such as specifications and usage history, is not fully observable from images alone. Consequently, the image-only model incurs a substantial CWC penalty despite producing moderately wide intervals.

The text-only model improves empirical coverage to a PICP of 0.945, reflecting the importance of textual descriptions that provide explicit specifications and usage details. However, the coverage remains marginally below the target level, resulting in a reduced but still non-negligible CWC penalty. This suggests that textual information alone, while more informative than images, is insufficient to fully capture the uncertainty present in RFP pricing.

In contrast, the proposed multimodal model consistently achieves a PICP of 0.958, exceeding the target confidence level and thereby eliminating the coverage penalty in CWC. At the same time, it produces substantially narrower intervals, with an MPIW of 121,700, compared to both unimodal baselines. As a result, the multimodal model attains the lowest CWC, demonstrating the most favorable balance between reliability and sharpness.

6.3 Impact of Category-wise Training

RFP categories differ substantially in price scale, variance, and distributional characteristics, ranging from low-cost items such as books to high-value assets such as cars and real estate. Training a single global pricing model across all categories requires simultaneously modeling these highly heterogeneous distributions, which can lead to scale dominance by high-value categories and unreliable uncertainty estimates for lower-priced products.

Category-wise training leads to more consistent calibration across categories, as reflected by PICP values that closely align with the target confidence level (Figure 3). In contrast, global models tend to exhibit under-coverage for low-priced categories and overly conservative intervals for high-priced categories, due to conflicting optimisation pressures across disparate price ranges. By isolating categories during training, the proposed approach avoids these effects and produces intervals that are both reliable and appropriately scaled.

6.4 Calibration and Reliability Analysis

Reliable uncertainty estimation is a central requirement for trustworthy pricing systems, particularly in RFP markets where price variability and incomplete information are inherent. Beyond producing narrow prediction intervals, a well-designed interval prediction model must achieve empirical coverage that closely matches the target confidence level.

The proposed framework demonstrates strong calibration across product categories, as reflected by PICP values that consistently align with the nominal 95% confidence level. This behavior indicates that the predicted intervals accurately capture the true price uncertainty rather than relying on overly conservative or overconfident estimates. The model avoids the common failure modes of interval predictors, such as under-coverage due to overconfident predictions or excessive interval width used to artificially inflate coverage.

Calibration quality is further supported by the observed relationship between PICP, MPIW, and CWC. Since the predicted intervals satisfy the coverage constraint across categories, the CWC metric reduces to MPIW, confirming that interval sharpness is achieved without sacrificing reliability. The category-wise variation in interval width reflects genuine differences in market uncertainty rather than calibration errors, reinforcing the interpretability of the predicted price.

6.5 Qualitative Results and Case Studies

To complement the quantitative evaluation, we present qualitative case studies illustrating the behavior of the proposed model on real RFP listings. Figure 4 shows representative

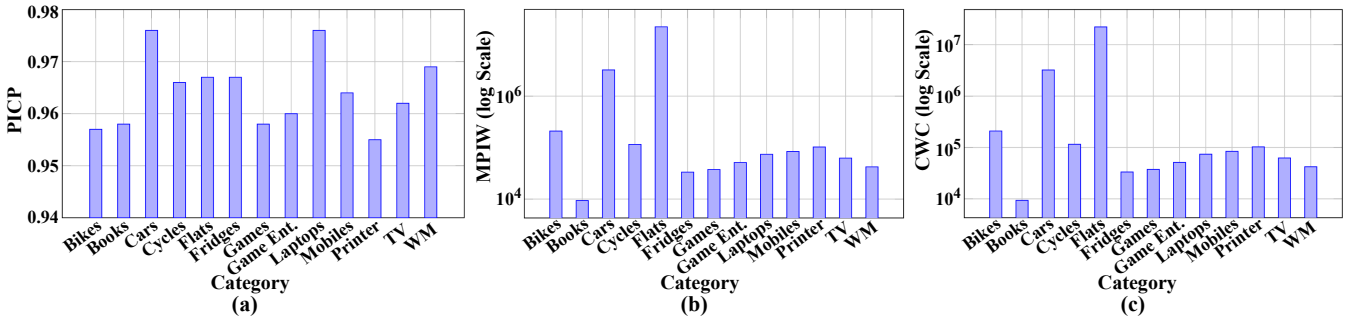

Figure 3: Category-wise evaluation of interval prediction performance. Subplots illustrate: (a) PICP, (b) MPIW (log scale), and (c) CWC (log scale). (Game Ent.= Game Entertainment, WM= Washing Machine)

Figure 4: Qualitative examples of predicted price intervals across diverse RFP categories. (AP= Actual Price, PIP= Pridicted Interval Price)

examples across diverse categories, including household appliances, consumer electronics, books, gaming accessories, and vehicles. For each example, the predicted price interval is displayed alongside the corresponding ground-truth resale price.

Across all shown cases, the predicted intervals successfully capture the true prices while adapting their width to the underlying price scale and category-specific uncertainty. Lower-priced items, such as books and accessories, are associated with relatively narrow intervals, whereas high-value products, including vehicles and laptops, result in wider intervals reflecting greater market variability. This adaptive behavior is consistent with the category-wise quantitative results and confirms that the model does not rely on uniform or overly conservative uncertainty estimates.

These qualitative examples also demonstrate the practical interpretability of the proposed uncertainty-aware pricing framework. By providing explicit price ranges rather than single point estimates, the model offers actionable information for both buyers and sellers, enabling risk-aware decision-making in real-world refurbished marketplaces. Importantly, the intervals remain informative while maintaining reliable coverage, reinforcing the calibration and robustness observed in the quantitative analysis.

6.6 Discussion

The results demonstrate that shifting from point estimation to interval prediction provides meaningful benefits in the context of RFP pricing. The combination of pretrained multimodal representations, feature-level fusion, category-aware modeling, and explicit interval optimization enables the proposed framework to deliver reliable and interpretable price ranges. These properties are essential for building trustworthy AI systems that support risk-aware decision-making in real-world RFP marketplaces.

7 Conclusion

This paper proposes a trustworthy multimodal interval-based framework for predicting refurbished product prices, explicitly addressing uncertainty arising from heterogeneous product conditions and incomplete multimodal information. By replacing point-wise estimates with calibrated price intervals, the approach provides more reliable and interpretable pricing guidance. Using pretrained CLIP-based image and text representations, multimodal fusion, and category-aware training, the framework achieves well-calibrated intervals with high coverage and reduced width across diverse product categories. Experiments on a large-scale dataset of 51,783 listings and qualitative case studies demonstrate that the proposed method supports risk-aware decision-making and enhances transparency in real-world refurbished product marketplaces.

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